HU Collection

TMUA Mock Set B Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	E	C	B	C	A	C	C	C	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	C	A	D	D	A	D	D	B	C

Each solution records the intended method and, where there is one, the trap the question was built around.

Question 1. Answer B.

Let $P = \cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$ and multiply by $\sin\frac{\pi}{7}$. Folding three times with $2\sin\theta\cos\theta = \sin 2\theta$,

$$\sin\tfrac{\pi}{7}\,P = \tfrac{1}{2}\sin\tfrac{2\pi}{7}\cos\tfrac{2\pi}{7}\cos\tfrac{4\pi}{7} = \tfrac{1}{4}\sin\tfrac{4\pi}{7}\cos\tfrac{4\pi}{7} = \tfrac{1}{8}\sin\tfrac{8\pi}{7} = -\tfrac{1}{8}\sin\tfrac{\pi}{7},$$

so $P = -\frac{1}{8}$. Since $\cos\frac{3\pi}{7} = -\cos\frac{4\pi}{7}$, the required product is $\frac{1}{8}$. Option E is P itself, the answer to the question that was not asked; the sign flip on the last cosine is the whole point.

Question 2. Answer E.

The cubic is $(x-2)^3 - 1$, so put $u = x - 2$; the limits become -3 and 3 and the integrand is $|u^3 - 1|$, which changes sign at $u = 1$. Then

$$\int_{-3}^{1}(1-u^3)\,du + \int_{1}^{3}(u^3-1)\,du = 24 + 18 = 42.$$

Ignoring the modulus gives $\int_{-3}^{3}(u^3-1)\,du = -6$, option A, which is negative and therefore impossible for the integral of a non-negative function. A negative answer to a modulus integral is always a signal to look again.

Question 3. Answer C.

The binomial coefficient $\binom{x}{j} = \frac{x(x-1)\cdots(x-j+1)}{j!}$ is a polynomial in x of degree j, and for a non-negative integer k we have $2^k = \sum_{j=0}^{k}\binom{k}{j}$, where the terms with $j > k$ vanish. So $p(x) = \sum_{j=0}^{4}\binom{x}{j}$ is a quartic with $p(k) = 2^k$ for $k = 0, \ldots, 4$, and it is the only one, two quartics agreeing at five points being equal. Then

$$p(5) = \sum_{j=0}^{4}\binom{5}{j} = 2^5 - \binom{5}{5} = 31.$$

The same answer comes from a difference table: the fourth differences of $1, 2, 4, 8, 16$ are constant, and extending the table one step gives 31. The trap is 32, which assumes a polynomial can imitate 2^x one step further than its degree allows.

Question 4. Answer B.

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Write $f(x) = x^3 - 3x + 1 = (x-a)(x-b)(x-c)$. Then $a^2 - 2 = (a-\sqrt{2})(a+\sqrt{2})$, and

$$\prod(a-\sqrt{2}) = -f(\sqrt{2}), \qquad \prod(a+\sqrt{2}) = -f(-\sqrt{2}),$$

so the required product is $f(\sqrt{2})f(-\sqrt{2}) = (1-\sqrt{2})(1+\sqrt{2}) = -1$. No root is ever found.

The trigonometric route explains the answer. Putting $x = 2\cos\theta$ turns $x^3 - 3x$ into $2\cos 3\theta$, so the roots are $2\cos\frac{2\pi}{9}$, $2\cos\frac{4\pi}{9}$, $2\cos\frac{8\pi}{9}$, and the map $x \mapsto x^2 - 2$ is $2\cos\theta \mapsto 2\cos 2\theta$, which permutes the three roots. The product is therefore $abc = -1$ by Vieta.

Question 5. Answer C.

Put $u = x + \frac{1}{x}$ and $v = y + \frac{1}{y}$. Then $u + v = 1 + \frac{1}{x} + \frac{1}{y} \geq 1 + \frac{4}{x+y} = 5$, the inequality being $\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$ (AM–HM, or clear denominators to $(x-y)^2 \geq 0$). Then

$$u^2 + v^2 \geq \frac{(u+v)^2}{2} \geq \frac{25}{2},$$

with equality throughout at $x = y = \frac{1}{2}$, where each bracket is $\frac{5}{2}$. Substituting $y = 1 - x$ and differentiating is possible but ugly, and the two-step inequality is what the question is testing.

Question 6. Answer A.

Expanding,

$$(\sin x + 2\cos x)(2\sin x + \cos x) = 2\sin^2 x + 5\sin x\cos x + 2\cos^2 x = 2 + 5\sin x\cos x.$$

Now $\sin x\cos x$ ranges over $[-\frac{1}{2}, \frac{1}{2}]$ (it equals $\frac{1}{2}\sin 2x$; or note $(\sin x \pm \cos x)^2 \geq 0$ gives $|\sin x\cos x| \leq \frac{1}{2}$ with both ends attained). So $M = \frac{9}{2}$, $m = -\frac{1}{2}$ and $Mm = -\frac{9}{4}$.

Question 7. Answer C.

Partial fractions in n alone go nowhere, because of the 2^n. The trick is to split the numerator against the factor 2 rather than against n and $n+1$: write $n + 2 = 2(n+1) - n$, so

$$\frac{n+2}{n(n+1)2^n} = \frac{2}{n\,2^n} - \frac{1}{(n+1)2^n} = \frac{1}{n\,2^{n-1}} - \frac{1}{(n+1)\,2^n}.$$

With $b_n = \dfrac{1}{n\,2^{n-1}}$ this is $b_n - b_{n+1}$, so the partial sum is $b_1 - b_{N+1} = 1 - \dfrac{1}{(N+1)2^N}$, and the infinite sum is 1. The first three partial sums are $\frac{3}{4}, \frac{11}{12}, \frac{31}{32}$, which is the pattern a candidate without the decomposition would have to guess from; option B is the first term on its own.

Question 8. Answer C.

Let $s = \sin x + \cos x$, so $\sin x\cos x = \frac{1}{2}(s^2 - 1)$. The equation becomes $s^2 - 1 = 2s - 2$, i.e. $(s-1)^2 = 0$, so $\sin x + \cos x = 1$. Squaring, $\sin x\cos x = 0$, which gives the candidates $0°, 90°, 180°, 270°, 360°$; but squaring is not reversible, and checking in $\sin x + \cos x = 1$ kills $180°$ and $270°$. The solutions are $0°, 90°, 360°$ with sum $450°$. Option F is the unchecked sum $900°$; option A forgets the endpoints.

Question 9. Answer C.

Label the heptagon $ABCDEFG$, with side 1, short diagonal p (two steps apart, e.g. AC) and long diagonal q (three steps apart, e.g. AD). Every regular polygon is cyclic, so Ptolemy applies to any

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four vertices. On the quadrilateral $ABCE$ the sides are $AB = BC = 1$, $CE = p$, $EA = q$ and the diagonals are $AC = p$, $BE = q$, so Ptolemy gives

$$AC \cdot BE = AB \cdot CE + BC \cdot EA, \qquad pq = p + q,$$

so $\frac{1}{p} + \frac{1}{q} = 1$ on dividing by pq. The lengths themselves are $p = 2\cos\frac{\pi}{7}$ and $q = 1 + p - \frac{1}{p}$, neither of which is needed. Ptolemy on $ABCD$ gives the companion identity $p^2 = 1 + q$.

Question 10. Answer A.

Since $\alpha\beta = -1$, we have $\frac{1}{\alpha} = -\beta$ and $\frac{1}{\beta} = -\alpha$, so the required value is $-(\alpha^5 + \beta^5)$. Writing $s_n = \alpha^n + \beta^n$ and using $s_{n+1} = 2s_n + s_{n-1}$ (from the equation), $s_1 = 2$, $s_2 = 6$, $s_3 = 14$, $s_4 = 34$, $s_5 = 82$. The answer is -82. Option F is the sign slip.

Question 11. Answer B.

The tangent to $y = x^2$ at $x = a$ is $y = 2ax - a^2$; the tangent to the second curve at $x = b$ is $y = (6 - 2b)x + b^2 - 10$. Matching gradients and intercepts, $a = 3 - b$ and $-(3 - b)^2 = b^2 - 10$, which reduces to $2b^2 - 6b - 1 = 0$. The two common tangents correspond to the two roots b_1, b_2, with gradients $6 - 2b_i$, so

$$(6 - 2b_1)(6 - 2b_2) = 36 - 12(b_1 + b_2) + 4b_1b_2 = 36 - 36 - 2 = -2$$

by Vieta, without solving for b at all. (The gradients are $3 \pm \sqrt{11}$.)

Question 12. Answer C.

The series is geometric with ratio $x - 1$; it converges exactly when $|x - 1| < 1$, i.e. $0 < x < 2$, and there its sum is $\dfrac{1}{1 - (x - 1)} = \dfrac{1}{2 - x}$. Outside that interval the formula $\frac{1}{2-x}$ still makes sense as a number but the series diverges, which is the trap in option A. At $x = 0$ the terms are ± 1 and the series does not converge, so option D is also wrong.

Question 13. Answer A.

A chord of length 6 in a circle of radius 5 sits at distance $\sqrt{25 - 9} = 4$ from the centre. Any line through $(3, 1)$ has distance at most $|OP| = \sqrt{10} < 4$ from the origin, since the perpendicular distance to a line through P cannot exceed the distance to P itself. So no such chord exists. The instinct to answer 2 works for chords of length 8, which need distance $3 < \sqrt{10}$.

Question 14. Answer D.

Compute a few terms: $a_1 = 3$, $a_2 = \frac{1}{2}$, $a_3 = -\frac{1}{3}$, $a_4 = -2$, $a_5 = 3$. The sequence has period 4. (Structurally, if $g(x) = \frac{x-1}{x+1}$ then $g(g(x)) = -\frac{1}{x}$, so g has order 4.) The product over one period is $3 \cdot \frac{1}{2} \cdot (-\frac{1}{3}) \cdot (-2) = 1$. Since $2026 = 4 \cdot 506 + 2$, the product is $a_1a_2 = \frac{3}{2}$.

Question 15. Answer D.

Let $t = 2^x + 2^{-x}$, so $t \geq 2$ by AM–GM, with $t = 2$ only at $x = 0$, and $4^x + 4^{-x} = t^2 - 2$. The equation becomes $t^2 - kt + 2 = 0$. Each root $t > 2$ gives two values of x (symmetric under $x \mapsto -x$), $t = 2$ gives one, and $t < 2$ gives none. The roots multiply to 2, so at most one of them can be ≥ 2. Hence exactly two solutions in x needs exactly one root $t > 2$, i.e. the quadratic $h(t) = t^2 - kt + 2$

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satisfies $h(2) < 0$, which is $6 - 2k < 0$, so $k > 3$. At $k = 3$ the roots are 2 and 1, giving only $x = 0$. Option B is the discriminant condition and ignores the constraint $t \geq 2$ entirely.

Question 16. Answer A.

$y' = 4x^3 - 12x^2 + 2ax = 2x(2x^2 - 6x + a)$. There is always the stationary point $x = 0$, so we need $2x^2 - 6x + a = 0$ to have no real roots: $36 - 8a < 0$, i.e. $a > \frac{9}{2}$. The boundary case $a = \frac{9}{2}$ gives the double root $x = \frac{3}{2}$, which is still a second stationary point (a stationary point of inflection), so option B is wrong. The factor $2x$ is the other trap: $x = 0$ is a stationary point for every a, so "no real roots" applies to the quadratic factor only.

Question 17. Answer D.

Split the integrand. The function $x\sqrt{4 - x^2}$ is odd, so its integral over $[-2, 2]$ is 0. What remains is

$$2\int_{-2}^{2} \sqrt{4 - x^2}\, dx,$$

and $y = \sqrt{4 - x^2}$ is the upper half of the circle of radius 2, so the integral is the area of that semicircle, 2π. The answer is 4π. There is no need for an antiderivative of $\sqrt{4 - x^2}$; recognising a definite integral as an area is the method. Option E takes the whole circle, option A forgets the even part.

Question 18. Answer D.

The angle at C is fixed, so C lies on an arc of a circle through A and B, and the sine rule in the circumcircle gives $2R = \dfrac{AB}{\sin 45^\circ} = \dfrac{4}{\sqrt{2}/2}$, so $R = 2\sqrt{2}$. The centre is at distance $\sqrt{R^2 - 2^2} = 2$ from AB. Since 45° is acute, C is on the major arc, on the far side of the centre from AB, and the largest height is $2 + 2\sqrt{2}$ at the midpoint of that arc. The maximum area is $\frac{1}{2} \cdot 4 \cdot (2 + 2\sqrt{2}) = 4 + 4\sqrt{2}$. Option A is the height, not the area.

Question 19. Answer B.

Substitute in each integral. In the first, $u = 2x$ gives $\frac{1}{2}\int_0^4 f = 3$, so $\int_0^4 f = 6$. In the second, $u = x + 1$ gives $\int_2^4 f = 5$. In the third, $u = 3 - 3x$ reverses the limits and contributes a factor $\frac{1}{3}$, giving $\frac{1}{3}\int_0^3 f = 2$, so $\int_0^3 f = 6$. Then

$$\int_3^4 f = \int_0^4 f - \int_0^3 f = 6 - 6 = 0.$$

The information is enough, so option G is wrong; the value 5 in option E is $\int_2^4 f$, read off carelessly.

Question 20. Answer C.

Do not try to invert f. Since f is increasing and $f(2) = 10$, the graph of f^{-1} on $[0, 10]$ is the reflection of the graph of f on $[0, 2]$ in the line $y = x$, and together the two regions fill the 2×10 rectangle:

$$\int_0^{10} f^{-1}(x)\, dx = 2 \cdot 10 - \int_0^2 (x^3 + x)\, dx = 20 - (4 + 2) = 14.$$

Option A is $\int_0^2 f$, the wrong piece of the rectangle.

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